

Geon Lee

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RESEARCH SUMMARY

My research goal is to uncover the fundamental principles by which interaction structures across relational data, retrieved knowledge, and multi-agent AI shape learning and emergent behavior. Building on these principles, I aim to develop principled frameworks for designing intelligent systems that can effectively harness complex interactions while remaining reliable, generalizable, and robust across diverse real-world environments.

WORK EXPERIENCE

University of Wisconsin-Madison

Postdoctoral Research Fellow (Advisor: Prof. Sharon Li)

Madison, WI, USA

Apr. 2027 (Expected) –

KAIST

Postdoctoral Research Fellow (Advisor: Prof. Kijung Shin)

Seoul, South Korea

Mar. 2026 – Mar. 2027 (Expected)

Snap Research

Research Intern (Mentors: Liam Collins, Clark Ju, Bhuvesh Kumar, Tong Zhao | Manager: Neil Shah)

Bellevue, WA, USA

Apr. – Jun. 2025

Data Augmentation for Generative Recommendation

NEC Labs America

Research Intern (Mentor: Wenchao Yu | Manager: Haifeng Chen)

Princeton, NJ, USA

May – Aug. 2023

Multi-Modal Time Series Forecasting

Amazon

Applied Scientist Intern (Mentor: Zhonghao Luo | Manager: Tao Ye)

San Francisco, CA, USA

Sep. – Dec. 2022

Conversational Music Recommendation for Alexa AI

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

M.S. & Ph.D. in Artificial Intelligence

Seoul, South Korea

Sep. 2020 – Feb. 2026

Advisor: Prof. Kijung Shin

PhD Dissertation Award, KAIST College of Engineering

Sungkyunkwan University (SKKU)

B.S. in Computer Science and Engineering

Suwon, South Korea

Mar. 2016 – Aug. 2019

GPA: 4.41/4.50; Major GPA: 4.45/4.50

Ranked 1st in the College of CSE

TUTORIALS

[1] Retrieval-Augmented Generation for LLM-Based Recommender Systems

Sunwoo Kim*, Geon Lee*, Kyungho Kim*, Liam Collins, Neil Shah, and Kijung Shin

CIKM 2026 & KDD 2026

[2] Mining of Real-World Hypergraphs: Patterns, Tools, and Generators

Geon Lee, Jaemin Yoo, and Kijung Shin

KDD 2023, WWW 2023, ICDM 2022, & CIKM 2022

PUBLICATIONS

- [1] Training-Free LLM-Based Recommendation with Post-LLM Item Refinement Using Collaborative Signals
Kyungho Kim, Sunwoo Kim, Geon Lee, Shinhwan Kang, Sojeong Kim, Liam Collins, Bhuvish Kumar, Donald Loveland, and Kijung Shin
CIKM 2026 (Short Paper)
- [2] Toward a Science of AI Agent Societies
Geon Lee*, Fanchen Bu*, Soo Yong Lee*, Sunwoo Kim, and Kijung Shin
KDD 2026 (Blue Sky)
- [3] Rethinking Contrastive Learning for Graph Collaborative Filtering: Limitations and A Simple Remedy
Geon Lee, Sunwoo Kim, Kyungho Kim, and Kijung Shin
ICML 2026
- [4] Hybrid-Vector Retrieval for Visually Rich Documents: Combining Single-Vector Efficiency and Multi-Vector Accuracy
Juyeon Kim, Geon Lee, Dongwon Choi, Taeuk Kim, and Kijung Shin
ACL 2026 (Findings)
- [5] ItemRAG: Item-Based Retrieval-Augmented Generation for LLM-Based Recommendation
Sunwoo Kim, Geon Lee, Kyungho Kim, Jaemin Yoo, and Kijung Shin
SIGIR 2026 (Short Paper)
- [6] ReFuGe: Feature Generation for Prediction Tasks on Relational Databases with LLM Agents
Kyungho Kim, Geon Lee, Juyeon Kim, Dongwon Choi, Shinhwan Kang, and Kijung Shin
WWW 2026 (Short Paper)
- [7] Sequential Data Augmentation for Generative Recommendation
Geon Lee, Bhuvish Kumar, Clark Mingxuan Ju, Tong Zhao, Kijung Shin, Neil Shah, and Liam Collins
WSDM 2026
- [8] Revisiting LightGCN: Unexpected Inflexibility, Inconsistency, and A Remedy Towards Improved Recommendation
Geon Lee, Kyungho Kim, Fanchen Bu, Langzhang Liang, and Kijung Shin
Transactions on Recommender Systems (2026) *Invited Extended Version of the RecSys 2024 Paper*
- [9] From Binary to General Attributes: Attributed Hypergraph Generation with Realistic Interplay Between Structure and Attributes
Jaewan Chun, Seokbum Yoon, Minyoung Choe, Geon Lee, and Kijung Shin
Knowledge and Information Systems (SCIE, 2026) *Invited Extended Version of the ICDM 2025 Paper*
- [10] A Comprehensive Survey on Multi-Behavior Recommender Systems: Extended Taxonomy and Recent Advances
Kyungho Kim, Sunwoo Kim, Geon Lee, Jinhong Jung, and Kijung Shin
International Journal of Data Science and Analytics (2026) *Invited to the PAKDD 2025 Special Issue*
- [11] RDB2G-Bench: A Comprehensive Benchmark for Automatic Graph Modeling of Relational Databases
Dongwon Choi, Sunwoo Kim, Juyeon Kim, Kyungho Kim, Geon Lee, Shinhwan Kang, Myunghwan Kim, and Kijung Shin
NeurIPS 2025 (Dataset & Benchmark)
- [12] TimeXL: Explainable Multi-modal Time Series Prediction with LLM-in-the-Loop
Yushan Jiang, Wenchao Yu, Geon Lee, Dongjin Song, Kijung Shin, Wei Cheng, Yanchi Liu, and Haifeng Chen
NeurIPS 2025
- [13] Attributed Hypergraph Generation with Realistic Interplay Between Structure and Attributes
Jaewan Chun, Seokbum Yoon, Minyoung Choe, Geon Lee, and Kijung Shin
ICDM 2025 *Best Paper Award; Best-Ranked Paper*

- [14] Identifying Group Anchors in Real-World Group Interactions Under Label Scarcity
Fanchen Bu, Geon Lee, Minyoung Choe, and Kijung Shin
ICDM 2025
- [15] A Self-Supervised Mixture-of-Experts Framework for Multi-behavior Recommendation
Kyungho Kim, Sunwoo Kim, Geon Lee, and Kijung Shin
CIKM 2025
- [16] SkySearch: Satellite Video Search at Scale
Minyoung Choe*, Geon Lee*, Changhun Han*, Suji Kim, Woong Hu, Hyebeen Hwang, Geunseok Park, Byeongyeon Kim, Hyesook Lee, Ha-Myung Park, and Kijung Shin
KDD 2025 (Applied Data Science)
- [17] KGMEEL: Knowledge Graph-Enhanced Multimodal Entity Linking
Juyeon Kim, Geon Lee, Taeuk Kim, and Kijung Shin
SIGIR 2025 (Short Paper)
- [18] MARIOH: Multiplicity-Aware Hypergraph Reconstruction
Kyuhan Lee, Geon Lee, and Kijung Shin
ICDE 2025
- [19] Multi-Behavior Recommender Systems: A Survey
Kyungho Kim, Sunwoo Kim, Geon Lee, Jinhong Jung, and Kijung Shin
PAKDD 2025 (Survey) *Best Survey Paper Award*
- [20] Beyond Neighbors: Distance-Generalized Graphlets for Enhanced Graph Characterization
Yeongho Kim, Yuyeong Kim, Geon Lee, and Kijung Shin
WWW 2025
- [21] TimeCAP: Learning to Contextualize, Augment, and Predict Time Series Events with Large Language Model Agents
Geon Lee, Wenchao Yu, Kijung Shin, Wei Cheng, and Haifeng Chen
AAAI 2025
- [22] A Survey on Hypergraph Mining: Patterns, Tools, and Generators
Geon Lee*, Fanchen Bu*, Tina Eliassi-Rad, and Kijung Shin
ACM Computing Surveys (SCI, 2025)
- [23] Resource2Box: Learning to Rank Resources in Distributed Search Using Box Embedding
Ulugbek Ergashev, Geon Lee, Kijung Shin, Eduard Dragut, and Weiyi Meng
ICDM 2024
- [24] Revisiting LightGCN: Unexpected Inflexibility, Inconsistency, and A Remedy Towards Improved Recommendation
Geon Lee, Kyungho Kim, and Kijung Shin
RecSys 2024 (Short Paper) *Best Short Paper Candidate*
- [25] Post-Training Embedding Enhancement for Long-Tail Recommendation
Geon Lee, Kyungho Kim, and Kijung Shin
CIKM 2024 (Short Paper)
- [26] Towards Better Utilization of Multiple Views for Bundle Recommendation
Kyungho Kim, Sunwoo Kim, Geon Lee, and Kijung Shin
CIKM 2024 (Short Paper)
- [27] Representative and Back-in-Time Sampling from Real-World Hypergraphs
Minyoung Choe, Jaemin Yoo, Geon Lee, Woonsung Baek, U Kang, and Kijung Shin
Transactions on Knowledge Discovery from Data (SCIE, 2024) *Invited Extended Version of the WWW 2022 Paper*

- [28] ViLLain: Self-Supervised Learning on Homogeneous Hypergraphs without Features via Virtual Label Propagation
Geon Lee, Soo Yong Lee, and Kijung Shin
WWW 2024
- [29] Random Walk with Restart on Hypergraphs: Fast Computation and an Application to Anomaly Detection
Jaewan Chun, Geon Lee, Kijung Shin, and Jinhong Jung
Data Mining and Knowledge Discovery (SCIE, 2023)
- [30] Hypergraph Motifs and Their Extensions Beyond Binary
Geon Lee*, Seokbum Yoon*, Jihoon Ko, Hyunju Kim, and Kijung Shin
The VLDB Journal (SCIE, 2023) *KAIST Outstanding Paper Award*
- [31] Hypercore Decomposition for Non-Fragile Hyperedges: Concepts, Algorithms, Observations, and Applications
Fanchen Bu, Geon Lee, and Kijung Shin
Data Mining and Knowledge Discovery (SCIE, 2023)
- [32] Temporal Hypergraph Motifs
Geon Lee and Kijung Shin
Knowledge and Information Systems (SCIE, 2023) *Invited Extended Version of the ICDM 2022 Paper*
- [33] Set2Box: Similarity Preserving Representation Learning for Sets
Geon Lee, Chanyoung Park, and Kijung Shin
ICDM 2022
- [34] HashNWalk: Hash and Random Walk Based Anomaly Detection in Hyperedge Streams
Geon Lee, Minyoung Choe, and Kijung Shin
IJCAI 2022
- [35] MiDaS: Representative Sampling from Real-World Hypergraphs
Minyoung Choe, Jaemin Yoo, Geon Lee, Woonsung Baek, U Kang, and Kijung Shin
WWW 2022
- [36] Simple Epidemic Models with Segmentation Can Be Better than Complex Ones
Geon Lee, Se-eun Yoon, and Kijung Shin
PLOS ONE (SCIE, 2022) *Oral, epiDAMIK @ KDD 2021*
- [37] THyMe+: Temporal Hypergraph Motifs and Fast Algorithms for Exact Counting
Geon Lee and Kijung Shin
ICDM 2021 *Best-Ranked Paper*
- [38] How Do Hyperedges Overlap in Real-World Hypergraphs? — Patterns, Measures, and Generators
Geon Lee*, Minyoung Choe*, and Kijung Shin
WWW 2021
- [39] Hypergraph Motifs: Concepts, Algorithms, and Discoveries
Geon Lee, Jihoon Ko, and Kijung Shin
VLDB 2020
- [40] MEGA: Multi-View Semi-Supervised Clustering of Hypergraphs
Joyce Jiyoung Whang, Rundong Du, Sangwon Jung, Geon Lee, Barry Drake, Qingqing Liu, Seonggoo Kang, and Haesun Park
VLDB 2020
- [41] Hyperlink Classification via Structured Graph Embedding
Geon Lee, Seonggoo Kang, and Joyce Jiyoung Whang
SIGIR 2019 (Short Paper)

ACADEMIC SERVICE

Conference Reviewer

• ACM SIGKDD (KDD)	2023 – 2027
• AAAI	2024 – 2027
• ACM CIKM	2022 – 2026
• Learning on Graphs (LoG)	2022 – 2026
• NeurIPS	2024 – 2026
• The Web Conference (WWW)	2024 – 2026
• ACM RecSys	2025 – 2026
• ICLR	2025 – 2026
• AISTATS	2025 – 2026
• ICML	2025 – 2026
• ACM MM	2025 – 2026
• ACML	2025 – 2026
• ARR	2026
• SDM	2024

Journal Reviewer

• IEEE Transactions on Knowledge and Data Engineering (TKDE)	2023 – 2026
• Information Sciences	2024 – 2026
• ACM Transactions on Information Systems (TOIS)	2025 – 2026
• IEEE Transactions on Network Science and Engineering (TNSE)	2024 – 2026
• IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	2025 – 2026
• Information Processing & Management	2025 – 2026
• Scientific Reports	2025 – 2026
• Machine Learning	2025 – 2026
• Neural Networks	2025 – 2026
• Patterns	2025 – 2026
• Transactions on Machine Learning Research (TMLR)	2026
• Journal of Machine Learning Research (JMLR)	2026
• Discover Artificial Intelligence	2026
• Expert Systems with Applications	2026
• Journal of Neuroscience Methods	2026
• Neurocomputing	2026
• Knowledge-Based Systems	2026
• IEEE Transactions on Neural Networks and Learning Systems (TNNLS)	2023 – 2025
• The VLDB Journal	2023 – 2025
• PLOS ONE	2024 – 2025
• Data Mining and Knowledge Discovery	2024 – 2025
• Big Data Research	2024

Session Chair

• ACM CIKM	2024
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AWARDS & HONORS

Gold Reviewer, ICML 2026	May 2026
PhD Dissertation Award, KAIST College of Engineering	Feb. 2026
Best Paper Award, ICDM 2025	Dec. 2025
One of the Best-Ranked Papers, ICDM 2025	Dec. 2025
Best Survey Paper Award, PAKDD 2025	Jun. 2025
Outstanding Reviewer, KDD 2025	Dec. 2024

KAIST Outstanding Paper Award	Nov. 2024
Best Short Paper Candidate , RecSys 2024	Oct. 2024
One of the Best-Ranked Papers , ICDM 2021	Dec. 2021
Sungkyunkwan Presidential Award	Aug. 2019
Sungkyunkwan Software Scholarship (Full tuition, all semesters)	2016 – 2019

TEACHING

Teaching Assistant

- KAIST AI506: Data Mining and Search Spring 2021, 2023
- KAIST AI607: Graph Mining and Social Network Analysis Fall 2020–2023
- KAIST AI617: Machine Learning for Robotics Spring 2022
- SKKU CSE3036: Seminar in Computer Engineering Fall 2019

Tutorial Presenter

- Mining of Real-World Hypergraphs: Patterns, Tools, and Generators CIKM 2022, ICDM 2022, WWW 2023, KDD 2023
- Retrieval-Augmented Generation for LLM-Based Recommender Systems KDD 2026, CIKM 2026

Guest Lecture

- Recent Advances in Recommender Systems (KAIST AI506) May 2026

TALKS

- Understanding Complex Interaction Systems with AI: From Human Networks to AI Agent Societies Yonsei University Mar. 2026